

Linear systems

Question 1 / 10 points / Sample student

Solve this system using row reduction. Show and label each row operation.

Write the final answer as a column vector.

$$2x + y = 5$$

$$x - y = 1$$

Submitted work

$$\left[\begin{array}{cc|c} 2 & 1 & 5 \\ 1 & -1 & 1 \end{array} \right] \quad \text{---->} \quad \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array} \right]$$

$$x = 2, y = 1$$

Proof by induction

Question 2 / 10 points / Sample student

Prove the following statement for every integer $n \geq 1$ by induction.

$$1 + 3 + \dots + (2n - 1) = n^2$$

State the inductive hypothesis for arbitrary $k \geq 1$ and finish with an explicit conclusion that the statement holds for every integer $n \geq 1$.

Submitted work

For $n = 1$, the sum is $1 = 1^2$.

Assume the claim is true for an arbitrary integer $k \geq 1$.

Then $1 + 3 + \dots + (2k - 1) = k^2$.

Adding the next odd number gives

$$k^2 + (2k + 1) = k^2 + 2k + 1 = (k + 1)^2.$$